Dohyun (Eric) Kim

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EDUCATION

University of Auckland - Auckland CBD, Auckland | Expected Graduation **Nov 2027**

- **Degree:** Bachelor's (Hons) of Engineering in Computer Systems Engineering
- **Concentration:** Embedded Systems & Intelligent Software
- **Related Coursework:** Computer Architecture · AI & Machine Learning · Hardware-Software Systems · Software Architecture · Digital Systems Design · Software Quality Assurance · OOP

PROJECTS

Sentiment PULSE - AWS | Individual Project | 2026

- Built a **serverless sentiment analysis pipeline** using AWS **Kinesis, Lambda, Comprehend, and DynamoDB** to demonstrate end-to-end cloud and AI engineering on AWS.
- Developed an event-driven NLP architecture with a live React dashboard using Kinesis-triggered Lambda and Comprehend sentiment classification **to showcase real-time data ingestion**, inference, and visualisation.

"Winnie the bot" - [3rd Place] ECSE Design Competition | Team Project | 2025

- Built **embedded hardware** for an AI-powered interactive robot using dual ATmega328P microcontrollers, servos, an AI camera, and audio peripherals to enable face tracking, arm movement, and voice dialogue.
- Contributed to **3D modelling and prototyping** of the robot enclosure using AutoCAD, **placing 3rd** in the UoA ECSE Design Competition.

Flappy Universe - VHDL | Team Project | 2026

- Implemented a **Flappy Bird-style game** in VHDL on an **Altera FPGA** with VGA signal generation, PS/2 mouse input, sprite rendering from ROM, and an LFSR random number generator.
- Developed a **layered graphics pipeline** using modular VHDL components and a pixel-priority compositing system to render animated game scenes at VGA resolution in real time.

Smart Energy Monitor - C & Embedded | Team Project | 2025

- Designed an embedded **energy monitoring system** using ATmega microcontrollers with sensor interfacing, ADC processing, and signal conditioning **to measure and display real-time energy usage**.
- Produced a validated PCB schematic using **Altium Designer and LTspice simulations** to achieve accurate sensor-to-display signal conditioning across the full hardware stack.

RoastWork Analytics - Python | Team Project | 2026

- Built a **desktop analytics application** using PyQt6, pandas, and Matplotlib to deliver **multi-segment sales dashboards with KPI tracking** across three business units.
- Implemented a **forecasting module** using Linear, Polynomial Regression, and Holt's Exponential Smoothing with scikit-learn, validated with MAE and RMSE holdout metrics.

EXPERIENCE

Creative Imaginary Lab (ciLab) - Robotics Instructor | Apr 2026 - Present

- Instructed students in robot **hardware assembly** and **software programming** to complete missions, while coaching and supporting teams in preparation for **nationwide robotics competitions**.

Twelve Restaurant - Front of House (FOH) | Jul 2024 - Jan 2025

- Delivered exceptional customer service and collaborated with colleagues to maintain smooth front of house operations, efficiently resolving complex customer situations.

SKILLS

Languages & Frameworks: Python, Java, C, JavaScript, R, VHDL, React.js, Pandas, Scikit-Learn

Tools & Platform: AWS (Lambda, Kinesis, DynamoDB, Comprehend), Git/GitHub, Android Studio, MATLAB, Altium Designer

LICENSES & CERTIFICATIONS

Amazon Web Services (AWS) Certified Cloud Practitioner | Apr 2026

- Demonstrated proficiency in **AWS core services** (EC2, S3, IAM, Lambda, VPC) and cloud architecture principles including the Well-Architected Framework, security best practices, and cost optimisation

Amazon Web Services (AWS) Certified AI Practitioner | May 2026

- Demonstrated foundational understanding of artificial intelligence, machine learning, and generative AI concepts, including practical application of **AWS AI/ML services** such as Amazon SageMaker, Bedrock, and Rekognition

ACTIVITIES AND LEADERSHIP

Institute of Electrical and Electronics Engineers (IEEE) | Jul 2025 **- Full-time Volunteer** @ NZ Robotics Olympiad 2025

Korean Engineering Body (KEB) | Jul 2024 - Present

- **Academic Team Executive -** Delivered tutorial sessions for junior engineering students and contributed to the planning of academic events supporting the student community.